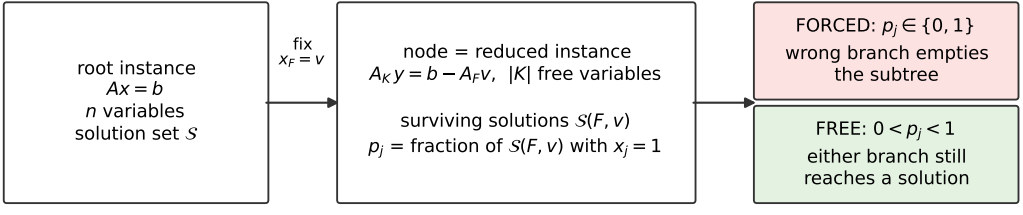
Conditioning on a partial assignment is the same as reducing the instance (Proposition 1)



Only errors on FORCED variables cost a backtrack. Their share is 3% at the root, 41% at depth 5 and 73% at depth 8 (10×25 in-tree states, $p = 0.3$), so the network is trained on reduced instances (in-tree states); labels are exact because $\mathcal{S}$ is enumerated.